Decoding of LDPC Codes Using Singularity of SNR Mismatch Over AWGN Channel

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***Abstract*—In this paper, we propose a new decoding algorithm that guarantees superior decoding performances for low-density parity-check (LDPC) codes without knowledge of channel parameters. Although obtaining correct channel parameters is very difficult in low-latency and energy-efficient communication systems, most studies on LDPC codes assume perfect knowledge of channel parameters. While receivers can obtain correct channel parameters by utilizing numerous pilot signals, increased overhead makes them infeasible in practical systems. In addition, using incorrect channel parameters for decoding usually degrades decoding performances. These challenges highlight the need for a new decoder that performs reliably without correct channel parameters. Meanwhile, the previous studies on repeat-accumulate (RA) codes and LDPC codes have observed that the signal-to-noise ratio (SNR) mismatch sometimes enhances decoding performances when the maximum number of decoding iterations is small. Inspired by these observations, we investigate singularity of the SNR mismatch in LDPC codes from numerous frame error rate (FER) simulations. By utilizing this singularity, we propose a new decoding algorithm for the energy-efficient communication system, where correct channel parameters are not available at receivers. The proposed decoder shows a performance gain of 0.25 dB at FER = 10^{-6} compared to the conventional belief propagation (BP) algorithm.**

***Index Terms*—Belief propagation (BP) algorithm, channel estimation error, energy-efficient decoder, low-density parity-check (LDPC) code, signal-to-noise ratio (SNR) mismatch.**

I. Introduction

Error-correcting codes (ECCs), which detect and correct errors during storage and transmission of information, are crucial technologies for data reliability in various areas including satellite communications, mobile networks, and data storage devices. Among them, low-density parity-check (LDPC) codes [1] are widely used in Wi-Fi and satellite broadcasting standards such as IEEE 802.11n [2], 802.11ac [3], and DVB-S2 [4] due to their superior error-correcting performances. In addition, LDPC codes used in 5G New Radio (NR) are designed with a rate-compatible structure, facilitating hardware implementation of encoding and allowing for flexibility in various data transmission rates and channel conditions [5]. Furthermore, LDPC codes have been researched for low-latency environments such as NR vehicle-to-everything (NR-V2X) and ultra-reliable low-latency communications (URLLC) [6], [7]. In the low-latency systems, the decoding latency is significantly influenced by both the codelength N and maximum number of decoding iterations $I_{\max}$. For example, the decoding latency increases linearly with $I_{\max}$ [8] and short packets of less than 1000 bits are preferred to satisfy URLLC latency and reliability constraints [9], [10]. Since low-latency and energy-efficiency are crucial in beyond 5G systems due to emerging applications such as artificial intelligence based Internet-of-Things (IoT), various decoding methods for ECCs have also been researched to ensure low-latency and energy-efficiency [11], [12].

The conventional decoding process of LDPC codes employs iterative message-passing based on the graph connection between variable nodes (VNs) and check nodes (CNs) to probabilistically estimate the value of each bit. For example, sum-product (SP) [13] and min-sum (MS) [14] algorithms are well-known decoding methods. Since LDPC codes are usually designed on the assumption that correct log-likelihood ratio (LLR) information is employed in decoding procedures, most studies on LDPC decoding assume perfect knowledge of channel parameters, which ensures superior decoding performances. However, it is very difficult for receivers to obtain the correct channel parameters (e.g., the noise distribution) because channel conditions usually vary each time data is transmitted in practical systems. Especially, limited resources in low-latency and energy-efficient communication systems result in channel estimation errors. Although increasing the number of pilot signals improves the accuracy of channel estimation, higher pilot overhead reduces the effective data transmission rate and requires more power [15], [16]. Moreover, channel estimation for each user is detrimental to the overall system latency in multi-user scenarios. In order to deal with these challenges, several studies have been researched for optimizing pilot overhead in URLLC and multi-user scenarios [17], [18], [19]. In [17], the pilot overhead was optimized for the given codelength, pilot length, and error probability. In addition, in [18], the reception quality and throughput were improved by estimating the channel and interference covariance matrix from reliable data symbols without increasing pilot length. From a multi-user point of view, [19]

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reduced pilot overhead by selecting users based on queue state information (QSI). Although pilot-based estimation has been widely researched for practical scenarios, it necessarily requires power allocation and induces high-latency due to the training duration. Instead, blind channel estimation that relies solely on received signals without pilot signals has been studied for low-latency and energy-efficient communication systems [20], [21], [22]. However, blind channel estimation is generally inaccurate compared to pilot-based channel estimation. Since estimation errors on channel parameters can result in performance degradation of LDPC codes in terms of frame error rates (FERs), some decoding methods with unknown channel parameters have been studied [23], [24], [25]. However, these decoding methods have some limitations that 1) decoding performances are severely degraded compared to the belief propagation (BP) algorithm with the correct channel parameters [23], [24] and 2) the very short codelength is assumed [25].

As practical and efficient decoding methods with unknown channel parameters, look-up-table (LUT) based decoders have been researched [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]. These decoders consist of message-passing tables for VNs and CNs using the finite number of alphabets. Although they are usually designed for a target channel parameter (e.g., E_b/N_0 over the additive white Gaussian noise (AWGN) channel), choosing an appropriate target parameter enables the superior decoding performance without knowledge of channel parameters [27], [28], [36]. Especially, some LUT based decoders that outperform the floating-point BP algorithm were proposed for regular LDPC codes with moderate N and small $I_{\max}$ [27], [28], [30]. Following those studies, designing LUT based decoders for irregular LDPC codes has been researched because the use of irregular LDPC codes is required in practical communication environments such as 5G NR [31], [32], [33], [34], [35], [36]. In contrast to regular LDPC codes that every VN/CN shares the same message-passing table, irregular LDPC codes require different tables for all VNs/CNs, which induces the high design complexity. Although the message alignment method was proposed to reduce the design complexity [31], LUT based decoders for irregular LDPC codes [31], [32], [33], [34], [35], [36] have not outperformed the floating-point BP algorithm with the correct channel parameters regardless of N and $I_{\max}$.

On the other hand, several studies have also been researched for the channel mismatch scenario in which receivers employ incorrect channel parameters for BP decoding. For example, FER performance degradation induced by the channel mismatch has been studied using Turbo codes [37], [38], [39], Luby-transform (LT) codes [40], and LDPC codes [41], [42], [43], [44]. In [43], the theoretical performance of LDPC codes has been analyzed based on the decoding threshold assuming the signal-to-noise ratio (SNR) mismatch over the AWGN channel. Although irregular LDPC codes that are robust to the SNR mismatch have also been designed based on the decoding thresholds, decoding performances deteriorate rapidly as the SNR mismatch is intensified. Meanwhile, the SNR mismatch for LDPC coded magnetic recording channels with partial-response equalization has been studied in [44]. Based on the density evolution analysis and simulations of BP decoding, it was observed that underestimation of the SNR (i.e., an estimated variance of the additive noise is larger than the correct value) sometimes improves the decoding performance of regular LDPC codes with moderate N and small $I_{\max}$. Similarly, the channel mismatch problem for repeat-accumulate (RA) codes has been studied to address SNR and carrier phase offset estimation errors [45]. In [45], it was also observed that the SNR mismatch sometimes improves the decoding performance of RA codes with short N and small $I_{\max}$. Although these observations occur for RA codes and regular LDPC codes, they show a possibility that irregular LDPC codes can be efficiently decoded without knowledge of channel parameters using the property of the SNR mismatch.

In this paper, we propose a new decoder for general LDPC codes that does not require knowledge of channel parameters (i.e., without using pilot signals) while guaranteeing the superior decoding performance. We first investigate singularity of the SNR mismatch in LDPC codes from numerous FER simulations. By utilizing singularity of the SNR mismatch and an estimation algorithm, we demonstrate that the proposed decoder outperforms the conventional BP algorithm with perfect knowledge of channel parameters for various types of LDPC codes and code-rates R when N is short and $I_{\max}$ is small. Thus, the proposed decoder is shown to be effective for low-latency and energy-efficient communication systems, where short N, small $I_{\max}$, and imperfect knowledge of channel parameters have to be considered. Our main contributions are summarized as follows:

- We analyze singularity of the SNR mismatch for BP decoding (i.e., SP and MS algorithms) of LDPC codes over the AWGN channel. From numerous simulation scenarios with various types of LDPC codes, R, N, and $I_{\max}$, we show that the decoding performance can be enhanced by inducing the SNR mismatch for some ranges of N and $I_{\max}$.
- Based on the aforementioned singularity and an estimation algorithm, we propose a new decoder for general LDPC codes that does not require knowledge of channel parameters. In particular, the proposed decoder is shown to be effective under practical coding scenarios with short N and small $I_{\max}$.
- Compared to the conventional BP algorithm with perfect knowledge of channel parameters, the proposed decoder achieves a performance gain of 0.25 dB at the FER of 10^{-6}, which highlights the effectiveness of the proposed decoding algorithm without knowledge of channel parameters.

This paper is organized as follows. In Section II, we describe the system model, basic properties of LDPC codes with the SNR mismatch, and some estimation algorithms. In Section III, we analyze singularity related to the SNR mismatch under various conditions. Then, we propose a new decoder by combining the channel estimation algorithm and singularity of the SNR mismatch in Section IV. Finally, Section V concludes the paper.

TABLE I
COMPARISON OF ESTIMATION ACCURACY BETWEEN MOM, MLE, AND EM ALGORITHMS

Channel condition	Algorithm	Average	Minimum	Maximum	Variance
SNR $= 0.1$ dB ($\sigma_n = 0.9886$)	MoM	0.97028	0.83801	1.10353	0.001915
	MLE	0.97204	0.83844	1.10359	0.001922
	EM	0.98747	0.8552	1.13263	0.002095
SNR $= 2.1$ dB ($\sigma_n = 0.7852$)	MoM	0.77928	0.67665	0.88542	0.001246
	MLE	0.78074	0.67701	0.88545	0.001252
	EM	0.78536	0.69038	0.88818	0.001177
SNR $= 5.1$ dB ($\sigma_n = 0.5559$)	MoM	0.554	0.48182	0.62939	0.00063
	MLE	0.55505	0.48206	0.62942	0.000633
	EM	0.55624	0.4942	0.62111	0.000416

II. BACKGROUND

A. System Model

In this paper, we consider LDPC codes with the short codelength ($N \leq 504$) and small maximum number of decoding iterations ($I_{\max} = 15$) for BP decoding. Then, we employ protograph-based Raptor-like (PBRL) codes [46] and 5G NR LDPC codes [5] that are known to be effective for short N and small $I_{\max}$. The AWGN channel and binary phase shift keying (BPSK) modulation are assumed, where the channel input x takes values of $\{-1, +1\}$. The channel output y is denoted by $y = x+n$, where $n \sim \mathcal{N}\left(0, \sigma_n^2\right)$ is the additive noise with the zero mean and variance σ_n^2. While many studies on LDPC codes typically assume perfect knowledge of channel parameters (i.e., receivers know σ_n^2) for decoding, we focus on the practical scenario, where the channel parameter is not given at the receiver. Note that this system model should be considered in low-latency and energy-efficient communication systems.

B. Properties of LDPC Codes With SNR Mismatch

If perfect knowledge of channel parameters is assumed over the AWGN channel, the distribution of channel LLRs follows $\mathcal{N}\left(\pm\frac{2}{\sigma_n^2}, \frac{4}{\sigma_n^2}\right)$, where the LLR value for the ith received signal y_i is $2y_i/\sigma_n^2$ ($i \in \{1, \ldots, N\}$). However, in the practical scenario with unknown channel parameters, the receiver determines the probability density function (PDF) of channel LLRs as $\mathcal{N}\left(\pm\frac{2}{\sigma_{es}^2}, \frac{4\eta}{\sigma_{es}^2}\right)$, where σ_{es}^2 is an estimated variance of the additive noise and $\eta = \sigma_n^2/\sigma_{es}^2$ is an SNR mismatch ratio. Then, the decoder calculates the LLR value for y_i as $2y_i/\sigma_{es}^2$. In the presence of the SNR mismatch, the PDF of channel LLRs does not satisfy the symmetry condition, which makes the use of the conventional extrinsic information transfer (EXIT) analysis [47], [48] infeasible. Therefore, the decoding threshold, which is the theoretical performance limit of LDPC codes, must be calculated through the density evolution [49] by computing probabilities for all messages during the iterative decoding. It is known that the case that $\sigma_{es}^2 > \sigma_n^2$ (under-estimated in terms of the SNR) is more detrimental than the case that $\sigma_{es}^2 < \sigma_n^2$ (over-estimated in terms of the SNR) for LDPC codes with the SNR mismatch [41], [42], [43]. However, for short N and small $I_{\max}$, we observe that the FER performance at the high SNR region is explicitly enhanced with the under-estimated SNR. Although this phenomenon cannot be explained from the theoretical analysis based on the decoding threshold, we demonstrate that this singularity generally holds for numerous cases in Section III-A.

C. Channel Estimation Algorithm

Since we assume the practical system with unknown channel parameters, it is necessary to estimate the channel parameters for use in the decoder. Thus, we first compare three well-known estimation algorithms: 1) moment of method (MoM) [50], 2) maximum likelihood estimation (MLE) [51], and 3) expectation-maximization (EM) [52] algorithms. In detail, we try to estimate σ_n using the aforementioned estimation algorithms and received signal y_i's with $N = 495$. The ith channel input x_i is randomly generated and Table I shows statistics for the estimated standard deviation of the additive noise σ_{es}.

We observe that average values of all estimation methods are close to the correct noise standard deviation in the region of the high SNR ($\sigma_n = 0.5559$). However, in the region of the low SNR ($\sigma_n = 0.9886$), average values from MoM and MLE algorithms deviate from σ_n whereas the average value of the EM algorithm is consistently close to σ_n. Although the EM algorithm seems to be effective in the average sense, the minimum and maximum of estimated values still significantly differ from σ_n. It indicates that the estimation algorithm is not sufficient for the practical scenario with unknown channel parameters. Thus, solely employing the estimation algorithm does not guarantee the superior decoding performance, which highlights the difficulty of the practical system that channel parameters are not given.

III. SINGULARITY OF SNR MISMATCH AND ANALYSIS

In this section, we describe singularity of the SNR mismatch for LDPC codes by evaluating FER results over the AWGN channel. Then, we analyze this phenomenon based on codelength and decoding iterations.

A. Singularity of SNR Mismatch

In the first two subsections, we introduce singularity of the SNR mismatch using the SP algorithm while considering various types of codelength, decoding iterations, LDPC codes, and code-rates. Then, we discuss the MS algorithm in the last subsection.

1) Codelength and Decoding Iterations: Based on PBRL codes lifted from the protograph P_1 in [46], we try to figure out singularity of the SNR mismatch in the perspective of codelength and decoding iterations. In Fig. 1, we compare FER

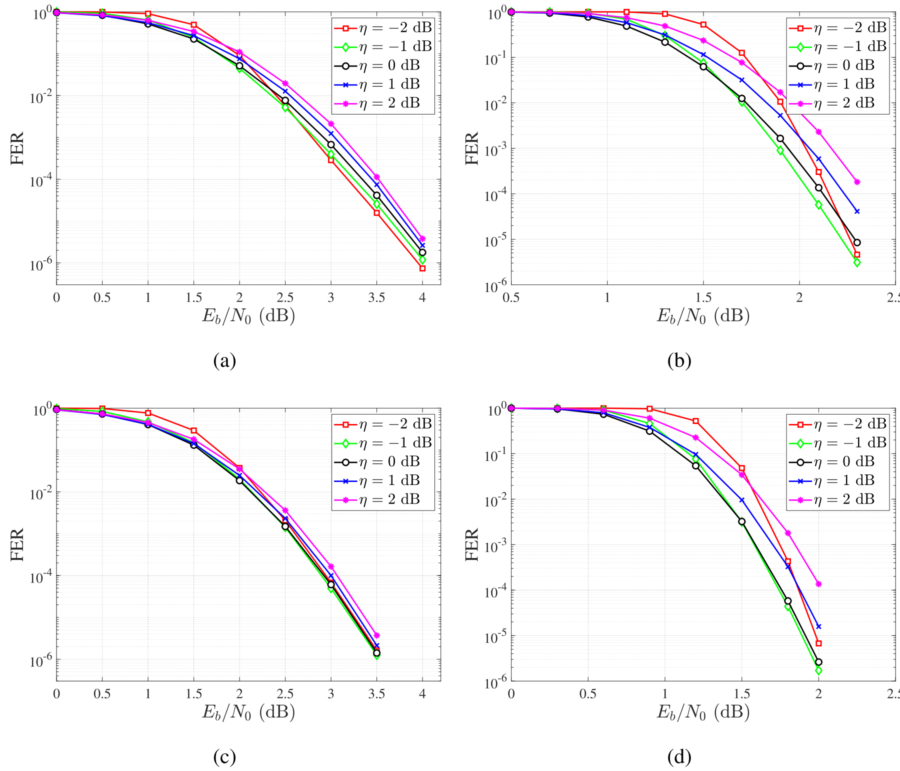


Fig. 1. FER performances with the SNR mismatch for LDPC codes lifted from P_1, where $R = 0.4$. (a): $N = 495$ and $I_{\max} = 15$. (b): $N = 2475$ and $I_{\max} = 15$. (c): $N = 495$ and $I_{\max} = 100$. (d): $N = 2475$ and $I_{\max} = 100$.

performances for the PBRL codes under four cases represented by $N = 495, 2475$ and $I_{\max} = 15, 100$. Assuming P_1 with the base size 9×15, lifting factors of 33 and 165 are employed. In each case of Fig. 1, the FER performances with positive values of η_{dB} are degraded compared to the performance with perfect knowledge of the channel parameter ($\eta_{\mathrm{dB}} = 0$), where η_{dB} is the decibel expression of η. This phenomenon is consistent with the previous studies [41], [42], [43]. However, we can confirm that the FER curves with negative values of η_{dB} outperform the FER curves with $\eta_{\mathrm{dB}} = 0$ as E_b/N_0 increases. In particular, the higher E_b/N_0, the better the FER curve with the low value of η_{dB}. Since this phenomenon is not expected for long N and large $I_{\max}$, where the FER performance is aligned with the theoretical metric such as the decoding threshold [41], [42], [43], we call it singularity of the SNR mismatch. Although we can observe this singularity for moderate N (2475) and $I_{\max}$ (100) as shown in Figs. 1(b)–1(d), the singularity in the practical case with short N (495) and small $I_{\max}$ (15) is much more noticeable as shown in Fig. 1(a). In particular, Fig. 1(d) shows that the singularity almost disappears when both N and $I_{\max}$ are moderate.

2) Various Codes and Code-Rates: In order to figure out whether singularity of the SNR mismatch generally holds for short N and small $I_{\max}$, we additionally evaluate FER performances across various protographs and code-rates in Fig. 2. We first describe FER results of LDPC codes lifted from P_1 with different code-rates (i.e., $R = 0.545$ and 0.75) in Figs. 2(a) and 2(b). Assuming two types of P_1 with the base sizes 5×11 and 2×8, lifting factors of 45 and 63 are employed. In addition to Fig. 1(a) ($R = 0.4$), we can confirm singularity of the SNR mismatch regardless of code-rates. Next, we present FER results of LDPC codes lifted from different protographs in Figs. 2(c) and 2(d). Assuming the protograph P_3 in [46] with the base size 10×16 and the 5G NR LDPC protograph in [5] with the base size 11×21, lifting factors of 33 and 32 are employed while considering the puncturing and shortening. Along with P_1, we again confirm singularity of the SNR mismatch regardless of protographs. Consequently, we have shown that FER performances can be enhanced with negative values of η_{dB} at the high E_b/N_0 region when N is short and $I_{\max}$ is small. Note that this singularity also occurs for RA codes and regular LDPC codes [44], [45], which supports our observation. Then, it is expected that we can utilize this singularity to enhance decoding performances of general LDPC codes with short N and small $I_{\max}$. In Section III-B, we further investigate how this singularity varies over the wider range of N and $I_{\max}$ in order to identify the target range in which singularity of the SNR mismatch becomes much more noticeable.

3) MS Algorithm: In addition to the SP algorithm, we employ the MS algorithm to demonstrate effects of the SNR mismatch. In Fig. 3, FER performances of LDPC codes lifted from P_1 are depicted using the MS algorithm. While short N and small $I_{\max}$ are assumed, FER curves are almost the same regardless of η_{dB}. In the previous studies [43], [53], it is shown that 1) the MS algorithm is robust to the SNR mismatch due to the simplified decoding operation of CNs

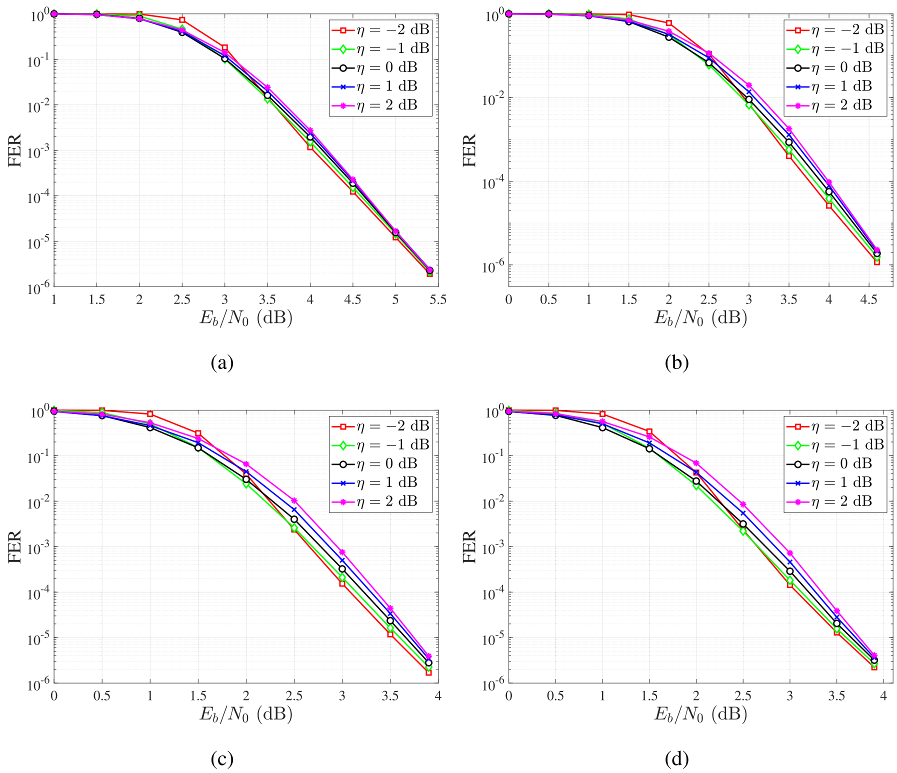


Fig. 2. FER performances with the SNR mismatch for LDPC codes lifted from various protographs, where $I_{\max} = 15$. (a): P_1 with $R = 0.545$ and $N = 495$. (b): P_1 with $R = 0.75$ and $N = 504$. (c): P_3 with $R = 0.4$ and $N = 495$. (d): 5G NR LDPC protograph with $R = 0.4$ and $N = 480$.

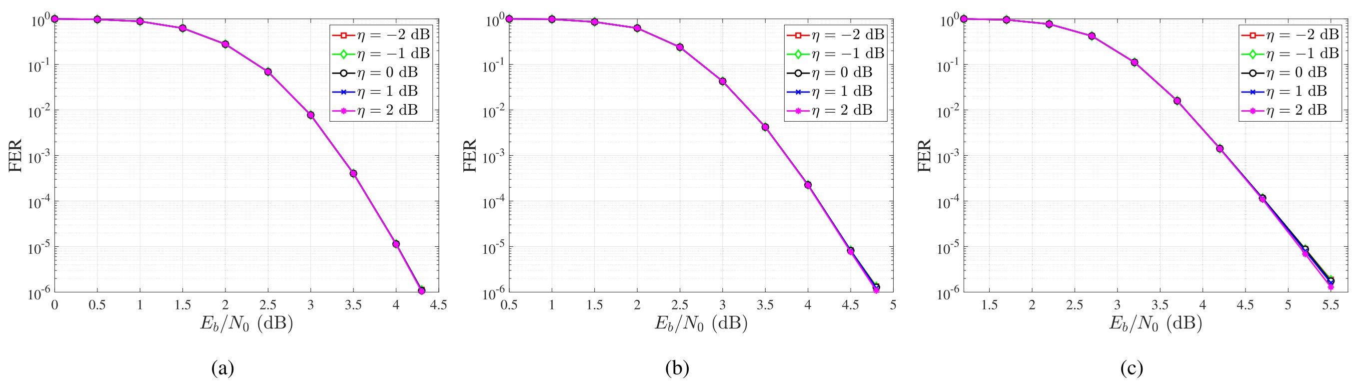
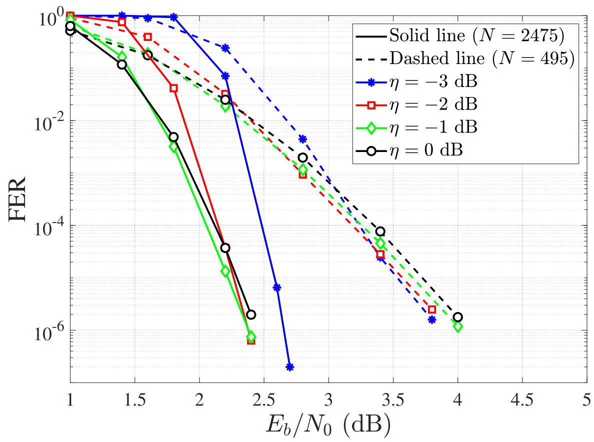


Fig. 3. FER performances with the SNR mismatch for LDPC codes lifted from P_1, where $I_{\max} = 15$ and the MS algorithm is applied. (a): $R = 0.4$ and $N = 495$. (b): $R = 0.545$ and $N = 495$. (c): $R = 0.75$ and $N = 504$.

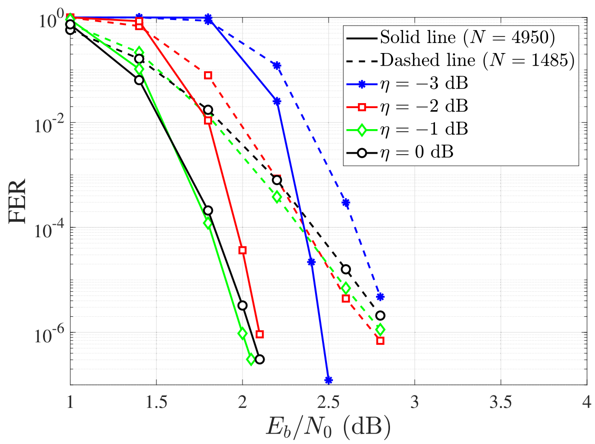
and 2) the same decoding threshold is achieved for all values of η_{dB} when the MS algorithm is adopted. Although these results assume long N and large $I_{\max}$, we can confirm that the same phenomenon holds for short N and small $I_{\max}$ in Fig. 3, which indicates that singularity of the SNR mismatch does not hold for the MS algorithm. Since the SP algorithm outperforms the MS algorithm (e.g., see Figs. 1(a) and 3(a)) and shows singularity of the SNR mismatch, we focus on the SP algorithm to enhance the decoding performance in this paper.

B. Analysis of Singularity Based on Codelength and Decoding Iterations

As shown in Fig. 1, singularity of the SNR mismatch is closely related to codelength and decoding iterations. In the perspective of codelength, the singularity is intensified for short N when the same $I_{\max}$ is assumed. Although this phenomenon cannot be explained from the decoding threshold, the decoding performance is also affected by the girth, minimum distance, and trapping set. These three metrics are closely related to each other, where short N degrades the girth and minimum distance [54], [55] while inducing harmful trapping sets [56], [57], [58], [59]. Especially, it is noted that the performance degradation due to the above metrics mainly occurs at the region of the high SNR, which can cause the error floor [56], [57], [60], [61]. Along with this, some post-processing methods have demonstrated that lowering or flipping channel LLRs is effective for dealing with the performance degradation [62], [63]. Since negative values of η_{dB} correspond to lowering channel LLRs, the singularity with short N (i.e., the performance improvement in the region of the high SNR when η_{dB} is negative)

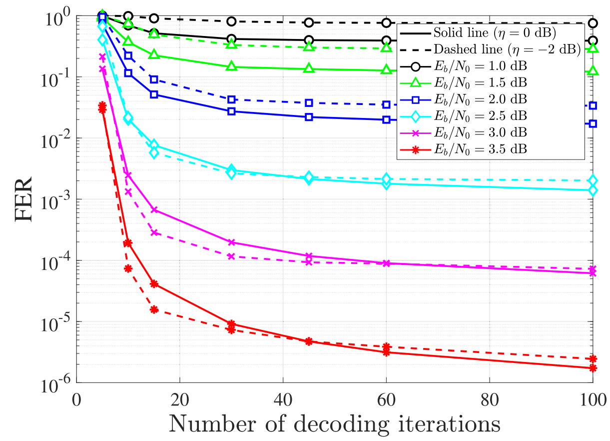


(a)

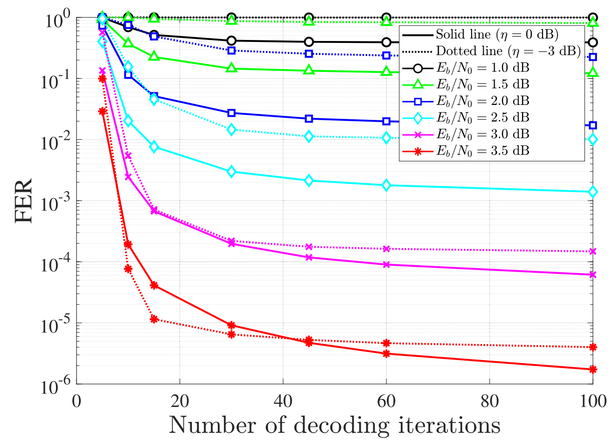


(b)

Fig. 4. Trends of FER performances based on the codelength, where $I_{\max} = 15$. LDPC codes are lifted from P_1, where $R = 0.4$. (a): $N = 495$ and 2475. (b): $N = 1485$ and 4950.



(a)



(b)

Fig. 5. Trends of FER performances based on the number of decoding iterations and E_b/N_0. The two types of η_{dB} are compared to $\eta_{\text{dB}} = 0$ and LDPC codes are lifted from P_1, where $R = 0.4$ and $N = 495$. (a): $\eta_{\text{dB}} = -2$. (b): $\eta_{\text{dB}} = -3$.

can also be expected from the previous studies [62], [63]. In general, the performance degradation induced by the aforementioned metrics is detrimental for small $I_{\max}$. Then, it is noted that singularity of the SNR mismatch is much more noticeable for small $I_{\max}$ (Fig. 1(a)) compared to moderate $I_{\max}$ (Fig. 1(c)). Since the PBRL codes [46] and 5G NR LDPC codes [5] are robust to the error floor, it does not explicitly occur for the simulated range of FERs as shown in Figs. 1 and 2. Nevertheless, the performance improvement observed at $\eta_{\text{dB}} < 0$ for short N can be interpreted in the similar manner as we can easily check that the performance improvement from the singularity also appears when the error floor exists.

In order to further investigate the effective range of N inducing the singularity, we compare FER performances of LDPC codes lifted from P_1 in Fig. 4, where $N = 495, 1485, 2475, 4950$. In the most cases, FER performances with $\eta_{\text{dB}} < 0$ outperform the FER performance with $\eta_{\text{dB}} = 0$ in the relatively high E_b/N_0 region. It is noted that the gain from the singularity is intensified as N decreases and the benefit of using $\eta_{\text{dB}} < 0$ becomes less significant as N increases. In particular, for $N = 4950$, the performance improvement from the singularity only occurs for $\eta_{\text{dB}} = -1$ whereas it occurs for $\eta_{\text{dB}} = -1$ and -2 when $N = 1485$ and 2475. Although the singularity is still present for relatively long N, its effect is much more noticeable for short N such as 495.

Next, we analyze the impact of decoding iterations on the singularity by discussing FER trends based on the number of decoding iterations and E_b/N_0 in Fig. 5. Although FER performances with $\eta_{\text{dB}} = -2$ and $\eta_{\text{dB}} = -3$ are degraded compared to the FER performance with $\eta_{\text{dB}} = 0$ for the most of decoding iterations and E_b/N_0, we can note that the singularity occurs when $I_{\max}$ is small and E_b/N_0 is high. For $\eta_{\text{dB}} = -2$, the performance gain over $\eta_{\text{dB}} = 0$ appears for $E_b/N_0 \geq 2.5$ dB and $10 \leq I_{\max} \leq 40$, where the gain is maximized around $I_{\max} = 15$. For $\eta_{\text{dB}} = -3$, the singularity does not appear at $E_b/N_0 = 2.5$ dB and 3.0 dB, unlike the case of $\eta_{\text{dB}} = -2$. However, the singularity occurs at $E_b/N_0 = 3.5$ dB for $10 \leq I_{\max} \leq 40$, where the performance gain around $I_{\max} = 15$ is larger than the gain obtained from $\eta_{\text{dB}} = -2$. Thus, the preferred value of η_{dB} is dependent on E_b/N_0 when $I_{\max}$ is small. In addition, decoding performances are improved rapidly in the region of small $I_{\max}$ as $I_{\max}$ increases. It comes from the fact that the decoding convergence requires at least some decoding iterations in the BP algorithm. Since channel LLRs affect decoding results crucially under the limited opportunities for error correction, lowering channel LLRs (i.e., negative values of η_{dB}) can be effective for the earlier decoding convergence when E_b/N_0 is high. From this point of view, the singularity with small $I_{\max}$ can also be expected. Thus, the performance improvement can be achieved by utilizing the singularity under short N and small $I_{\max}$. In Section IV, a new decoder that utilizes the singularity together with an estimation algorithm is proposed to guarantee the superior decoding performance without knowledge of channel parameters.

IV. Proposed Decoding Algorithm and Simulation Results

In Section III, we analyzed singularity of the SNR mismatch under short N and small $I_{\max}$. In particular, FER curves with negative values of η_{dB} intersect at certain E_b/N_0 points, which we refer to as crossover points. From a new decoder that utilizes singularity of the SNR mismatch, we demonstrate that the decoding performance can be improved without knowledge of channel parameters. First, we generate probability mass functions (PMFs) of estimated channel parameters at crossover points, where the EM algorithm is employed. Next, we propose a new decoding algorithm that adjusts channel LLRs based on the iteration number. Finally, we show that the proposed decoding algorithm outperforms the conventional BP algorithm when N is short and $I_{\max}$ is small.

A. Generation of PMFs for Estimated Channel Parameters

In order to determine crossover points, we first evaluate N_p FER graphs with distinct values of η_{dB}, which are denoted by $\eta_{\mathrm{dB},k}$, $k \in \{1, \cdots, N_p\}$. For selecting $\eta_{\mathrm{dB},k}$, we assume that $\eta_{\mathrm{dB},k-1} > \eta_{\mathrm{dB},k}$ and $\eta_{\mathrm{dB},1} = 0$. For example, $N_p = 4$ and $\eta_{\mathrm{dB,k}} \in \{0, -1, -2, -3\}$ in Fig. 6. Then, we define the crossover point E_b/N_{0k} as the point in which the two FER curves with $\eta_{\mathrm{dB},k}$ and $\eta_{\mathrm{dB},k-1}$ intersect. In addition, we denote the corresponding standard deviation of the additive noise by σ_{n_k}. For $k = 1$, we set $E_b/N_{01} = E_b/N_{0\mathrm{th}}$, where $E_b/N_{0\mathrm{th}}$ is the decoding threshold of a given protograph. For instance, crossover points are depicted in Fig. 6. After crossover points are determined, we generate PMFs for estimated channel parameters at the crossover points using the EM algorithm. By obtaining 10^6 estimated channel parameters $\hat{\sigma}$'s at each crossover point E_b/N_{0k} from the EM algorithm and received signal (i.e., the same procedure in Section II-C), we make the distribution of them as $\mathrm{PMF}_k(\hat{\sigma})$ by forming a histogram with 1024 uniform intervals over the observed range. All estimated channel parameters that belong to the same interval are regarded as the center value of that interval. By counting and normalizing the number of samples in each interval, we obtain $\mathrm{PMF}_k(\hat{\sigma})$. Note that these PMFs are generated only once and subsequently reused. They are not regenerated during the decoding procedure.

Before explaining the necessity of $\mathrm{PMF}_k(\hat{\sigma})$, we discuss how to utilize singularity of the SNR mismatch. Since the FER performance with $\eta_{\mathrm{dB},k}$ outperforms the FER performance with

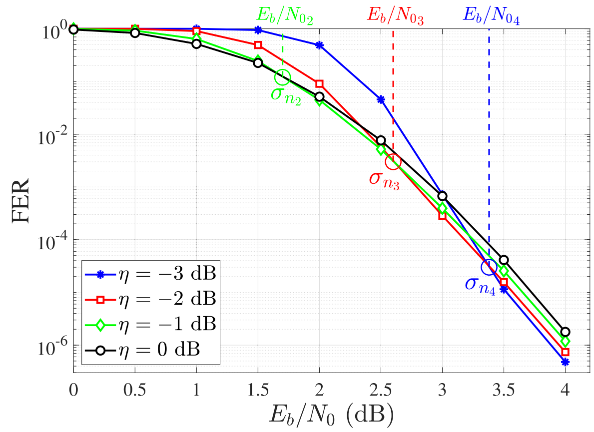


Fig. 6. An example for crossover points of the LDPC code lifted from P_1, where $N_p = 4$, $\eta_{\mathrm{dB},k} \in \{0, -1, -2, -3\}$, $R = 0.4$, $N = 495$, and $I_{\max} = 15$.

$\eta_{\mathrm{dB},k-1}$ for $E_b/N_0 \geq E_b/N_{0k}$, we can improve the FER performance by effectively adjusting channel LLRs (i.e., adjusting η_{dB}). Specifically, we can enhance the decoding performance by selecting the superior value of η_{dB} at each E_b/N_0 point if we have perfect knowledge of channel parameters (i.e., if we know the correct value of E_b/N_0). For example, it is beneficial to employ σ_{es_k} for calculating channel LLRs at the crossover point E_b/N_{0k}, where σ_{es_k} is determined as follows:

$$\sigma_{es_k} = \frac{\sigma_{n_k}}{\sqrt{10^{\frac{\eta_{\mathrm{dB,k}}}{10}}}}.$$

Since knowledge of channel parameters is not given in our scenario, we have to estimate the channel parameter from the received signal and EM algorithm in order to utilize singularity of the SNR mismatch. However, the estimated channel parameter is often inaccurate as shown in Table I. To compensate for this inaccuracy, we try to derive the similarity between the estimated channel parameter and crossover points rather than solely employing the estimated channel parameter. Although $\mathrm{PMF}_k(\hat{\sigma})$ represents the probability distribution of estimated values at E_b/N_{0k}, it can be used to determine which σ_{n_k} is the most similar to the estimated channel parameter $\hat{\sigma}$. In the next subsection, we propose a decoding algorithm that effectively utilizes singularity of the SNR mismatch from $\mathrm{PMF}_k(\hat{\sigma})$ and σ_{es_k}.

B. Proposed Decoding Algorithm

In this subsection, we propose a new decoding algorithm that effectively utilizes singularity of the SNR mismatch without knowledge of channel parameters. As mentioned in Section IV-A, we focus on calculating the similarity between $\hat{\sigma}$ and each σ_{n_k} rather than finding the correct value of σ_n. Then, we determine which σ_{es_k} is employed for calculating channel LLRs for each decoding iteration. This approach is motivated by the fact that $\hat{\sigma}$ obtained from the EM algorithm often deviates from the correct value of σ_n as shown in Table I. Consequently, utilizing a single fixed σ_{es} determined by $\hat{\sigma}$ and FER graphs with different values of η_{dB} for all decoding iterations may lead to performance degradation. Therefore, the

TABLE II
SUMMARY OF NOTATIONS USED IN ALG. 1

Notation	Description	Related line number
I	The current decoding iteration number	1, 12, 13, 22
EM(y_i's)	An estimated standard deviation of the additive noise for the received signal y_i's from the EM algorithm	2
$\{l_1, l_2, \cdots, l_{N_p}\}$	A set of sorted indices for $\{1, 2, \cdots, N_p\}$ in the descending order based on $\{w_1, w_2, \cdots, w_{N_p}\}$	11, 13, 15, 19
$\mathrm{LLR}_{ch,i}$	A channel LLR of the ith VN	15, 19
q	An index satisfying $\sum_{t=1}^{q-1} w_{l_t} \leq I < \sum_{t=1}^{q} w_{l_t}$	19
LLR_i^{app}	A posteriori probability (APP) LLR of the ith VN	24
d_i	A decoded bit for the ith VN	24, 25, 26
c_i	A transmitted bit for the ith VN	26

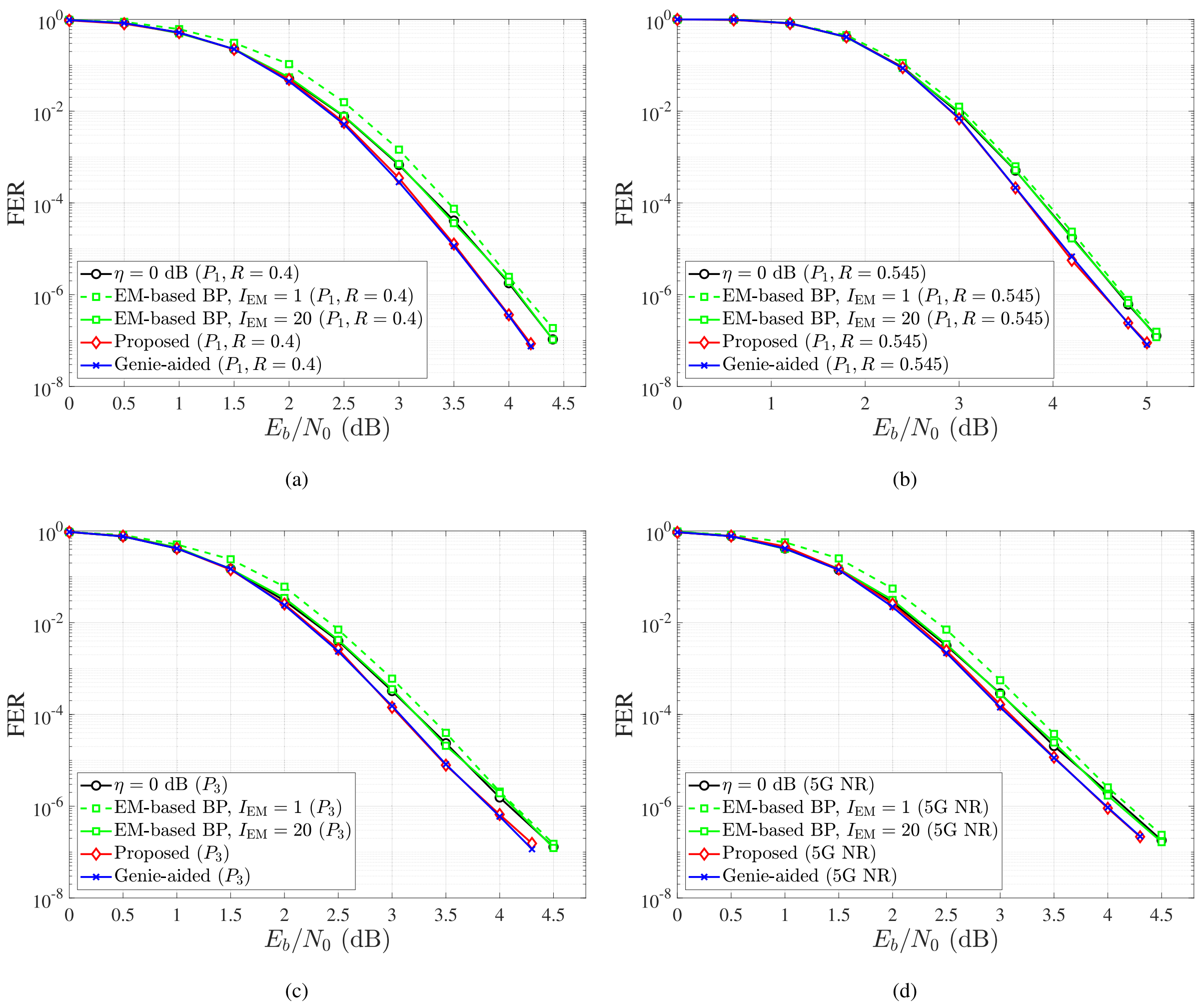


Fig. 7. Comparison of FER performances for LDPC codes under various protographs and code-rates, where $I_{\max} = 15$. (a): P_1 with $R = 0.4$ and $N = 495$. (b): P_1 with $R = 0.545$ and $N = 495$. (c): P_3 with $R = 0.4$ and $N = 495$. (d): 5G NR LDPC protograph with $R = 0.4$ and $N = 480$.

proposed decoding algorithm is designed to select σ_{es_k} for channel LLR calculation at each decoding iteration based on the similarity between $\hat{\sigma}$ and each σ_{n_k}. Since only N_p types of σ_{es_k} are considered, the decoder utilizes the mismatch ratio that deviates from $\eta_{\mathrm{dB},k}$'s. However, we show that it is enough for a performance improvement in Section IV-C. The proposed algorithm is described in Alg. 1 and the related notations are summarized in Table II. First, the proposed decoding algorithm employs the EM algorithm to estimate the channel parameter $\hat{\sigma}$ and then computes its similarity to σ_{n_k} as $p_k = \mathrm{PMF}_k(\hat{\sigma})$. Next, the number of decoding iterations w_k is determined in proportion to p_k, where σ_{es_k} is employed for generating channel LLRs. Based on the similarity, σ_{es_k}'s are sorted in the descending order and then they are sequentially employed for generating channel LLRs. In the below, we provide an example for the proposed decoding algorithm.

Example 1: Let $N_p = 4$, $I_{\max} = 15$, and $\{w_1, w_2, w_3, w_4\} = \{7, 2, 4, 1\}$ from line 7 of Alg. 1. Then, $\{l_1, l_2, l_3, l_4\} = \{1, 3, 2, 4\}$

Algorithm 1 Proposed Decoding Algorithm Utilizing Singularity of the SNR Mismatch

Input: Parity check matrix H, y_i's, $I_{\max}$, N_p, $\mathrm{PMF}_k(\hat{\sigma})$, σ_{es_k}
Output: Decoding result φ

1: $I \leftarrow 0$, $w \leftarrow 0$, $\varphi \leftarrow 0$.
2: $\hat{\sigma} \leftarrow \mathrm{EM}(y_i\text{'s})$.
3: **for** $k = 1 : N_p$ **do**
4: $\quad p_k \leftarrow \mathrm{PMF}_k(\hat{\sigma})$.
5: **end for**
6: **for** $k = 1 : N_p$ **do**
7: $\quad w_k \leftarrow \left\lfloor I_{\max} \cdot p_k / \sum_{k=1}^{N_p} p_k \right\rfloor$.
8: $\quad w \leftarrow w + w_k$.
9: **end for**
10: $w \leftarrow I_{\max} - w$.
11: $w_{l_1} \leftarrow w_{l_1} + w$.
12: **while** $I < I_{\max}$ **do**
13: $\quad$ **if** $I < w_{l_1}$ **then**
14: $\quad\quad$ **for** $i = 1 : N$ **do**
15: $\quad\quad\quad \mathrm{LLR}_{ch,i} = 2y_i/\sigma^2_{es_{l_1}}$.
16: $\quad\quad$ **end for**
17: $\quad$ **else**
18: $\quad\quad$ **for** $i = 1 : N$ **do**
19: $\quad\quad\quad \mathrm{LLR}_{ch,i} = 2y_i/\sigma^2_{es_{l_q}}$.
20: $\quad\quad$ **end for**
21: $\quad$ **end if**
22: $\quad I \leftarrow I + 1$.
23: $\quad$ **Message passing update:** Perform CN to VN, VN to CN, and APP LLR updates using the SP algorithm [13].
24: $\quad$ **APP LLR evaluation:** Based on LLR_i^{app}, determine d_i as follows:

$$d_i = \begin{cases} 0, & \text{if} \quad \mathrm{LLR}_i^{app} > 0 \\ 1, & \text{otherwise} \end{cases}.$$

25: $\quad$ **if** $H \cdot (d_1, d_2, \cdots, d_N)^T = \mathbf{0} \pmod 2$ **then**
26: $\quad\quad$ **if** $d_i = c_i$, $\forall i$ **then**
27: $\quad\quad\quad \varphi \leftarrow 1$, go to **line** 33.
28: $\quad\quad$ **else**
29: $\quad\quad\quad$ Go to **line** 33.
30: $\quad\quad$ **end if**
31: $\quad$ **end if**
32: **end while**
33: **return** φ

and w_1 is adjusted to 8 from line 11 of Alg. 1. For the current decoding iteration number I, the following σ_{es_k} is employed for generating channel LLRs:

- σ_{es_1} for $I < 8$.
- σ_{es_3} for $8 \leq I < 12$.
- σ_{es_2} for $12 \leq I < 14$.
- σ_{es_4} for $14 \leq I < I_{\max}$.

Note that extrinsic messages of VNs and CNs are not initialized although channel LLRs are adjusted.

C. FER Results

In the proposed decoding algorithm, selection of N_p and $\eta_{\mathrm{dB},k}$'s is the most important for a performance improvement. From the Monte Carlo simulation, we determine $N_p = 4$ and $\eta_{\mathrm{dB},k} \in \{0, -1, -2, -3\}$ as in Fig. 6. For various types of protographs and code-rates (i.e., P_1, P_3 in [46], 5G NR protograph in [5], and $R = 0.4, 0.545$), FER performances of the proposed decoding algorithm are given in Fig. 7. For a comparison, we also evaluate FER performances for four cases: 1) the conventional BP algorithm with perfect knowledge of channel parameters (denoted by $\eta = 0$ dB), 2) the conventional BP algorithm with channel parameters obtained from the EM algorithm, where the number of EM iterations I_{EM} is set to 1 (denoted by EM-based BP, $I_{\mathrm{EM}} = 1$), 3) the conventional BP algorithm with channel parameters obtained from the EM algorithm, where I_{EM} is set to 20 (denoted by EM-based BP, $I_{\mathrm{EM}} = 20$), and 4) the genie-aided decoder that utilizes the superior value of η_{dB} at each E_b/N_0 point (denoted by Genie-aided). For example, η_{dB} for the genie-aided decoder is determined as follows:

$$\eta_{\mathrm{dB}} = \begin{cases} 0, & E_b/N_0 < E_b/N_{02} \\ -1, & E_b/N_{02} \leq E_b/N_0 < E_b/N_{03} \\ -2, & E_b/N_{03} \leq E_b/N_0 < E_b/N_{04} \\ -3, & E_b/N_0 \geq E_b/N_{04} \end{cases}.$$

In Fig. 7(a), the proposed decoding algorithm achieves a performance gain of 0.25 dB at the FER of 10^{-6} compared to the conventional BP algorithms with perfect knowledge of channel parameters and with channel parameters obtained from the EM algorithm. This result highlights the effectiveness of the proposed decoding algorithm for short N and small $I_{\max}$ because it does not require any knowledge of channel parameters. In this comparison, the EM-based BP algorithm can be regarded as a practical evaluation baseline that represents decoding without knowledge of channel parameters. In particular, we consider two extreme EM-based BP cases under $I_{\mathrm{EM}} = 1$ and 20 (i.e., iterations utilized only for estimation in the EM algorithm). Since the estimation of the EM algorithm is inaccurate with $I_{\mathrm{EM}} = 1$, we can note that the decoding performance is degraded for $I_{\mathrm{EM}} = 1$. In contrast, if I_{EM} is set to the relatively large number of 20 to ensure the maximum performance based on only the channel estimation algorithm in practical scenarios, we can obtain the almost identical FER curve to that of the ideal BP decoder with $\eta = 0$ dB. Despite this, it still does not outperform the proposed decoding algorithm, which supports the superiority of the proposed decoding algorithm in practical scenarios.

Furthermore, the proposed decoding algorithm shows the almost same decoding performance to the genie-aided decoder. It indicates that 1) N_p types of σ_{es_k} are sufficient to utilize singularity of the SNR mismatch and 2) assigning decoding iterations based on the similarity is an efficient strategy. The proposed decoding algorithm dynamically adjusts σ_{es} based on the estimated channel parameter from the EM algorithm and the pre-determined set of $\mathrm{PMF}_k(\hat{\sigma})$ to calculate channel LLRs. In contrast to the genie-aided decoder that relies on a single fixed σ_{es}, the proposed decoding algorithm employs N_p types of σ_{es_k} during the decoding procedure. By sequentially using multiple σ_{es_k}'s, it provides diversity in LLR scaling, where the previous studies [12], [64], [65] have demonstrated that LLR scaling and dynamic scheduling at each decoding iteration usually provide superior decoding performances. In addition, the proposed decoding algorithm does not initialize extrinsic messages of VNs and CNs but updates them cumulatively although σ_{es} is changed. For these reasons, the proposed decoding algorithm can achieve the almost same decoding performance to the genie-aided decoder. These results can also be confirmed regardless of protographs and code-rates as shown in Figs. 7(b)–7(d). Consequently, the proposed decoding algorithm is effective in the practical scenario with short N and small $I_{\max}$, where channel parameters are not available at the receiver. Note that this scenario is closely related to low-latency and energy-efficient communication systems, which highlights the importance of the proposed decoding algorithm. Although the proposed decoding algorithm requires the pre-processing with the estimation algorithm, it is explicitly better than employing pilot signals in the perspective of energy-efficiency.

V. Conclusion

In this paper, we analyzed singularity of the SNR mismatch for LDPC codes. Through extensive simulations, we confirmed that the SNR mismatch sometimes provides a performance improvement. In particular, this phenomenon is much more noticeable when the codelength is short and maximum number of decoding iterations is small. Based on the singularity and channel estimation algorithm, we proposed a new decoding algorithm that ensures the superior decoding performance without knowledge of channel parameters. To utilize singularity of the SNR mismatch, we generated PMFs of estimated channel parameters employing the EM algorithm. Based on the PMFs and EM algorithm, the proposed decoding algorithm adjusts channel LLRs to employ singularity of the SNR mismatch during the decoding process.

From numerous FER simulations, we demonstrated that the proposed decoding algorithm achieves the superior decoding performance compared to the conventional BP algorithm with perfect knowledge of channel parameters. Thus, we can conclude that the proposed decoding algorithm is effective for the practical scenario because it does not require any knowledge of channel parameters. For future work, LDPC codes can be optimized for the proposed decoding algorithm. For example, we expect that optimization based on the singularity, mismatch threshold, and minimum distance can be conducted while considering characteristics of the proposed decoding algorithm.

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